Impact of Local Time on St. Patrick Geomagnetic Storm response at mid-low latitudes

Castellanos-Velazco C. I.[1,3] Corona-Romero P.[2,3] Sergeeva M.[2,3]
Orrala-Legorreta I. D Orrala-Legorreta I.D.[1,3], Gonzalez-Esparza J. A.[2,3]

[1]Universidad Nacional Autonoma de Mexico, Posgrado en Ciencias de la Tierra
[2] Laboratorio Nacional de Clima Espacial
[3] Instituto de Geofisica

Abstract

This study analyzes magnetic fluctuations associated with Prompt Penetration Electric Fields (PPEFs) using data from four mid-low latitude observatories separated by 60°–90° in longitude, covering different local time sectors. We focus on the St. Patrick's Day geomagnetic storm of 17 March 2015. Using Fourier analysis, we reconstruct the magnetic signatures of ionospheric currents. Our results demonstrate that these ionospheric magnetic responses depend strongly on longitudinal location during the storm's main phase and are particularly influenced by partial ring current activity.

Keywords: Regional Space weather, Magnetic fluctuations, Prompt-Penetration Electric Fields

1 Introduction

A geomagnetic storm (GS) is a temporary disturbance of Earth's magnetosphere caused by the injection of energy and charged particles into the system, leading to a significant reduction in the geomagnetic field intensity. These GS are typically triggered by solar phenomena such as Coronal Mass Ejections (CMEs) or high-speed solar wind (HSSW) streams. Due to their potential impacts on technological systems—including disruptions to communications, navigation, and power grids—GSs are among the most extensively studied phenomena in space weather research.

During the Main Phase (MP) of a GS, energy and charged particles are transported into Earth's inner magnetosphere. Since the magnetosphere is dynamically coupled with the ionosphere, particularly at high latitudes [C.T Russell, 2016], the ionosphere exhibits pronounced responses to these perturbations. One critical ionospheric effect is the generation of storm-time electric fields, which can be manifested at different latitudes:

- Prompt Penetration Electric Fields (PPEF) [Nishida, 1968a,b]
- Disturbed Dynamo Electric Fields (DDEF) [Blanc and Richmond, 1980]

The concept of PPEF was first introduced by Nishida [1968a,b], who observed periodic magnetic fluctuations between 4 hours and 30 minutes in ground magnetometers located around the auroral oval and the dip equator, both showing nearly identical fluctuations. Later, Vasyliunas [1970], Jaggi and Wolf [1973] developed the first model explaining the penetration of convective electric fields ($\mathbf{E_C}$) during disturbed periods, propagating through Region 1 Field-Aligned Currents (R1-FAC). During quiet periods, $\mathbf{E_C}$ is shielded by a polarized electric field induced by Region 2 Field-Aligned Currents (R2-FAC). However, during the MP of a GS, magnetic reconnection leads to an almost immediate intensification of R1-FAC activity due to the injection of solar wind particles and energy. Since R2-FAC depends on ring current activity to strengthen, it initially fails the shielding of $\mathbf{E_C}$, allowing its transfer to the ionosphere through R1-FAC. This process continues until R2-FAC intensifies its activity though the intensification of ring current, allowing R2-FAC balancing R1-FAC activity, which can take from tens of minutes to several hours [Wei et al., 2015]. During this period of imbalance, $\mathbf{E_C}$ penetrates into the mid- and low-latitude ionosphere, being that the reason to be called Penetration Electric fields. Furthermore, Kelley et al.

[1979], Wei et al. [2015] showed that PPEF depends on the southward orientation of the interplanetary magnetic field (IMF) component B_Z and can also be triggered by sharp directional changes in B_Z.

DDEF on the other hand, are associated with an ionospheric current driven by the injection of charged particles into the auroral ionospher. This injection, produces a Joule heating and leads to the circulation of equatorward thermospheric winds which drag charged particles (protons) Blanc and Richmond [1980]. With the dragged protons, it gets to a charge separation which induces a Pedersen current in the same direction of neutral wind flow. Near the equator, the Coriolis effect produces proton accumulation and westward circulation occur around the dip equator, generating a poleward electric field. Consequently, a second Pedersen current is induced in the poleward direction, opposing the first current. At mid-latitudes ($\sim 40°$), the above commented phenomena, drives Hall currents, which are interrupted at the terminators (limit of day-side with night-side) and get a closure or circuit at adjacent latitudes. This process results in vortex-shaped currents known as the Disturbed Dynamo, characterized by directions opposite to the Solar Quiet (SQ) current. For this reason, the disturbed dynamo is also referred to as the anti-SQ current.

At mid-low latitudes ($20 - 30°$), geomagnetic storms can significantly affect electron density through two primary mechanisms: Prompt Penetration Electric Fields (PPEF) and disturbance dynamo-driven neutral winds. On the dayside, an eastward PPEF causes an upward $\mathbf{E} \times \mathbf{B}$ drift, lifting the plasma to higher altitudes where recombination is slower, often leading to a positive ionospheric storm (TEC increase). In contrast, the storm-induced circulation from high to low latitudes transports molecular-rich air ($[O_2]$, $[N_2]$), increasing the $\frac{[N_2]}{[O]}$ ratio. This change in atmospheric composition enhances the loss rate of atomic oxygen ions (O^+) through recombination, which can lead to a negative ionospheric storm (TEC decrease) [Vijaya Lekshmi et al., 2011, Prölss, 1995].

Both PPEF and DDEF were studied by Fejer et al. [1983], who observed their effects using incoherent scatter radar at Jicamarca latitude. Later, Amory-Mazaudier [1982] detected DDEF above Saint-Santin latitude, with results consistent with the disturbed dynamo model. On the other hand, Le Huy and Amory-Mazaudier [2005] modeled by the first time the magnetic field associated with ionospheric disturbances, assuming that, for mid-low latitudes, they are mainly driven by DDEF and PPEF. They considered that during the MP, PPEF magnetic effects are dominant, whereas during the recovery phase (RP), regional magnetic field variations are mainly driven by DDEF. Following this initial approach, several studies [Le Huy and Amory-Mazaudier, 2008, Zaka et al., 2009] have focused on analyzing local responses at different latitudes and longitudes to analyze the different local responses associate to DDEF phenomena. In Liu et al. [2013], a statistical study was conducted to understand ionospheric responses as a function of local sectors during GS onset. Later, Younas et al. [2020] proposed using Fourier analysis to isolate magnetic fluctuations related to DDEF and PPEF based on their quasi-periodic behavior, identifying periods of approximately 24 hours for DDEF and less than 4 hours for PPEF.

In case of Mexico, regional space weather has a relatively recent history. On 2014, it was form the Mexican Space Weather Service (SCIESMEX), leading to combined efforts to understand the regional space weather phenomena, in the middle of a global perspective Gonzalez-Esparza et al. [2017]. From this teamwork, studies like Corona-Romero et al. [2018, 2017] has been focused on local geomagnetic response during GS periods, and the computing of a local K index specific for Mexico. On the other hand, Romero-Hernandez et al. [2017], Sergeeva et al. [2017], Sergeeva et al. [2020] found important ionospheric activity related to geomagnetic storms. In Castellanos-Velazco et al. [2024], the main focus was analyzing the local geomagnetic response related to PPEF and DDEF phenomena, during 20 strong GS, from 2003 to 2018, where one of the main conclusions were the need of comparing the geomagnetic response in the center of Mexico with different locations at similar geomagnetic latitudes.

In this paper, we analyze the PPEF magnetic signatures at different longitude sectors, looking for the sector where these fluctuations are more intense. We compare the results with the Asymmetric Ring Current activity in order to analyze it's influence on PPEF strength and so, on local geomagnetic response. We selected the GS of march 17^{th}, 2015 which is also known as the St. Patrick's event. This GS is known to be the most intense GS during the solar cycle 24^{th}, reaching -274 nT during its MP.

2 St. Patrick Geomagnetic Storm

The geomagnetic storm (GS) analyzed in this work is the intense event that occurred on March 17, 2015, widely known as the St. Patrick's Day storm. It is considered the most intense storm of solar cycle 24 Wu et al. [2016]. The storm was triggered by a coronal mass ejection (CME) associated with a C9.1 solar flare and a subsequent dimming observed on March 15. This CME, recorded as a partial halo by the SOHO/LASCO coronagraphs, had an estimated speed of approximately 660 km/s.

Figure 1 provides an overview of the interplanetary conditions that led to the St. Patrick's Day storm. A more detailed description about the trigger event is provided by Wu et al. [2016].

Panel (a) shows the total interplanetary magnetic field, B_T (black line), and its north-south component, B_Z (red line). Panels (b) and (c) presents solar wind properties, including the flow speed V_X (positive in anti sunward direction, black line), proton density n_P (orange line) and proton temperature T_P (blue line). In panel (b) also shows the driven Interplanetary Electric Field E_Y, calculated as $-V_X \times B_Z$. Finally, panel (e) displays the geomagnetic response through the SYMH index (green line).

The key milestones of the event are marked by three vertical dashed lines in Figure 1. The first line (left) indicates the arrival of an interplanetary shock (IPS) at 04:30 UT at the L1 point. The IPS signature is clear from the step-like increases in B_T, V_X and n_P. This shock compressed the magnetosphere, causing a Sudden Storm Commencement (SSC) visible as a sharp increase in the SYMH index between 04:30 and 07:00 UT, highlighted by light gray shading.

The region following the shock, the CME sheath, is identified by highly fluctuating B_T and B_Z, enhanced n_P and rising T_P. A period of southward B_Z between 06:00 and 09:00 UT initiated the what we called, the first stage of main phase (MP), driving the SYMH index down to its first minimum of -80 nT.

The second dashed line (14:00 UT) marks the arrival of the magnetic cloud, identified by Wu et al. [2016] as the main driver of the GS. Its passage is identified by a smooth, strong rotation of the magnetic field, a drop in temperature, and so, a corresponding low plasma beta. A sustained and intense southward B_Z component (reaching -40 nT) within the cloud marked the beginning of the second (and more intense) stage of the MP. The SYMH index decreased sharply, reaching its global minimum of -234 nT at 22:47 UT. The entire main phase period is shaded in dark gray in panel (e).

The third dashed line (23:00 UT) indicates the trailing edge of the magnetic cloud and the entry into a high-speed solar wind (HSSW) stream. While the high solar wind speed persists, the geomagnetic activity subsides as B_Z turns northward and the E_Y drops close to zero, ending the geoeffective driving of the storm.

3 Procedure

3.1 Geomagnetic Data

The geomagnetic data analyzed in this paper were downloaded from the INTERMAGNET public database (*International Real-Time Magnetic Observatory Network* [INTERMAGNET, 2021]), corresponding to the observatories (OB) listed in Table 1. All OB are located within a range of magnetic latitude (18°–33°) and are longitudinally separated by 60°–90° from each other. The closest OB is KAK, while the farthest is TEO from JAI. However, this spacing was tolerated since KAK is the closest OB to the longitude of 165° E.

To study how the local time (LT) sector affects the geomagnetic response during the main phase (MP) of a geomagnetic storm (GS), we focused on data from the four observatories (OB) listed in Table 1. Our aim was to analyze the GS response at these OB during a time window (TW) of maximum activity from the partial ring current (PRC), from 14:07 to 01:00 UT (11 hours), covering the second stage of the MP and the first few hours of the recovery phase (RP).

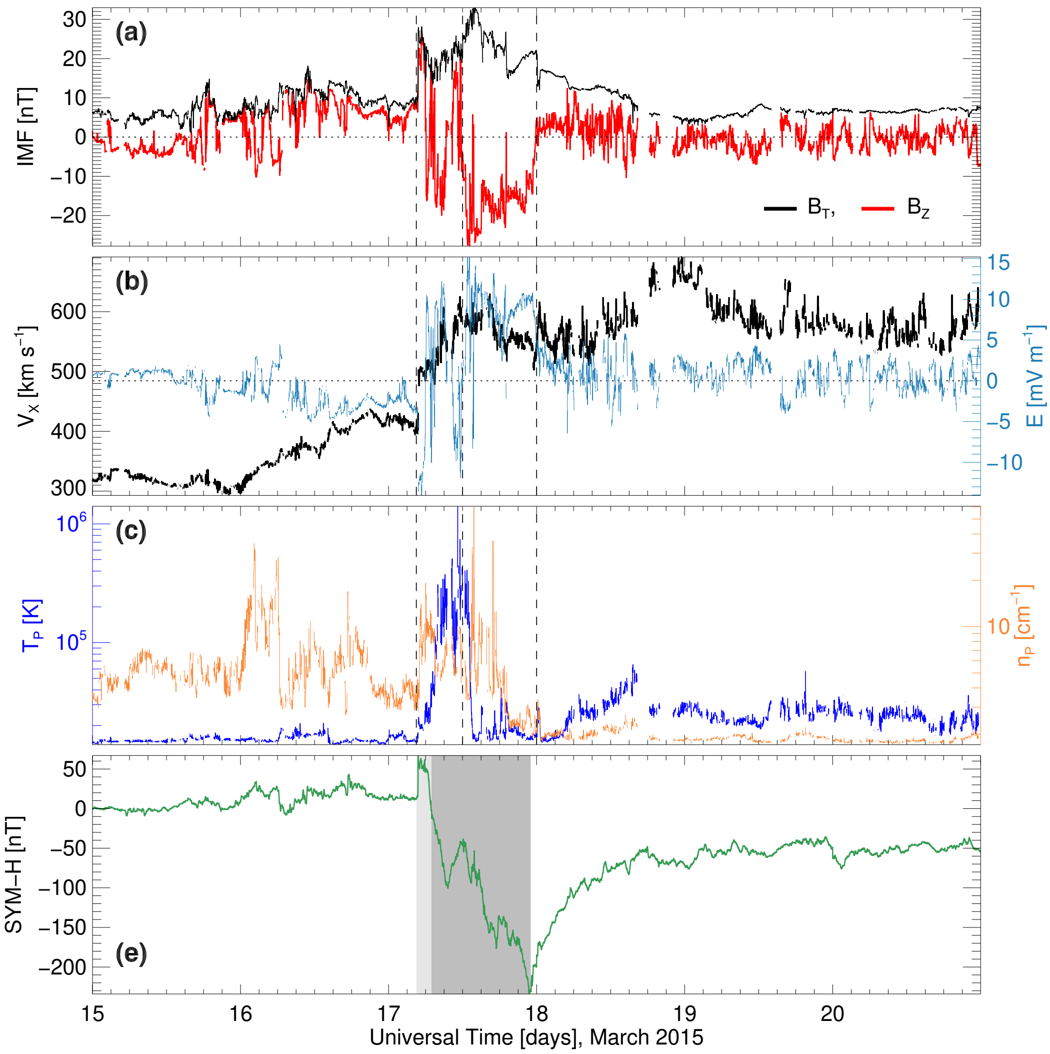

Figure 1: Interplanetary measurements. Panel (a): interplanetary magnetic field total component (black line) and z component (red line). Panel (b): V_X component of solar wind velocity (black line) and Interplanetary Electric Field (blue line). Panel (c): Proton Temperature (blue line) and Proton Density (orange line). Panel (d): SYMH (green line) and ASYMH index (orange line), highlighting the initial phase (light gray shade), main phase 1 (medium gray shade) and main phase 2 (dark gray shade).

Table 1: List of observatories considered for this study. Each OB is separated longitudinally at $70-90°$ apart.

Observatory	IAGA code	Country/ Territory	Latitude		Longitude	
			Geographic	Magnetic	Geographic	Magnetic
Guimar	GUI	Spain	28.317 N	33.21 N	16.433 W	61.1 E
Jaipur	JAI	India	26.917 N	18.42 N	75.8 E	150.47 E
Kakioka	KAK	Japan	36.232 N	27.8 N	140.186 E	149.73 W
Teoloyucan	TEO	Mexico	19.747 N	27.81 N	99.182 W	28.4 W

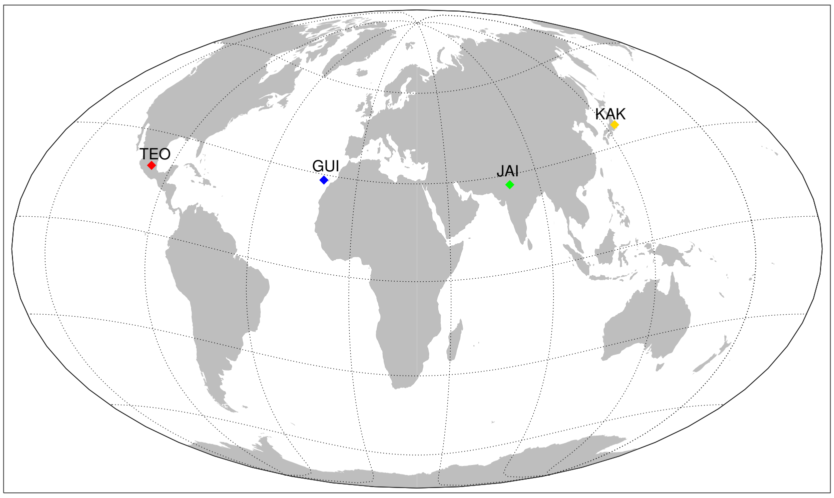

Figure 2: Geographic position of the four magnetic OB, distributed at $70 - 90°$ from each other

We therefore considered four local TW sectors during which the MP of the GS was active. These sectors are:

- Dawn-Noon (DWN) sector, 06:01 - 12:00 LT.
- Noon-Dusk (NDS) sector, 12:01 - 18:00 LT.
- Dusk-Midnight (DSM) sector, 18:01 - 00:00 LT.
- Midnight-Dawn (MDW) sector, 00:01 - 06:00 LT

Considering that the analyzed TW lasts approximately 11 hours and the local times at which it begins and ends, each OB is located within approximately two of these sectors during the TW. For this reason, we divided the TW into two subsets (TW1 and TW2), which for each OB cover two of the mentioned LT sectors (covering most of the sectors for GUI and KAK).

In Table 2, we present a brief description of the time windows during which each OB was located in TW1 and TW2. It can be seen that TEO was located in the DWN and NDS sectors, while JAI was located in the DSM and MDW sectors, placing these two at almost longitudinally opposite locations. GUI and KAK, on the other hand, are slightly shifted from the NDS/DSM and MDW/DWN sectors, respectively. This is consistent with the fact that they are not exactly $90° \pm 10°$ apart from the adjacent OB.

Table 2: Characteristics of the St. Patrick's Day event. Column 1 displays TW1 and TW2 in UT spanning March 17 and the first hour of March 18. Columns 2 and 3 present the minimum (SYMH) and maximum (ASYH) values observed during the main phase, respectively. Columns 4–7 show the local time sectors for each observatory during TW1 and TW2.

GS TW	Min SYMH	Max ASYH	Local TW			
UT	[nT]	[nT]	GUI	JAI	KAK	TEO
TW1 13:01 - 19:00	-234	273	11:01 - 17:00	18:01 - 00:00	22:01 - 04:00	06:01 - 12:00
TW2 19:00 - 01:00	-234	174	17:00 - 23:00	00:00 - 06:00	04:00 - 10:00	12:00 - 18:00

3.2 Magnetic fluctuations of ionospheric origin

It is well accepted that [Rishbeth and Garriott, 1969, Knecht and B.M, 1985, Gjerloev, 2012, van de Kamp, 2013] local recordings H_{loc} of a ground magnetometer can be considered a linear combination of different geomagnetic sources:

$$H_{loc} = H_{SQ} + H_0 + H_{MT} + H_{others} + H_I, \quad (1)$$

where H_{SQ} represents the diurnal variation and H_0 is the main field during a month window. H_{MT} represents the variations induced by the intensification of magnetospheric currents during the GS. In the case of H_{others}, it corresponds to magnetic sources less significant, like (and no limited to)) magnetotellurics, crustal magnetization, etc; which can be neglected for not being considered as relevant compared to the others mentioned. Finally, H_I corresponds to the magnetic variations driven by ionospheric current activity during a GS. Thus, the local geomagnetic response during a GS is given by:

$$\Delta H = H_{loc} - (H_{SQ} + H_0). \quad (2)$$

We provide a detailed explanation of how we model the diurnal variation and the monthly baseline in Appendix A.

From Le Huy and Amory-Mazaudier [2005], Amory-Mazaudier et al. [2017], Younas et al. [2020], for mid-low magnetic latitudes, with λ as the geomagnetic latitude of each OB, we can approximate H_{MT} as:

$$H_{MT} \approx SYMH \cdot \cos(\lambda), \quad (3)$$

Thus, H_I can be approximated as:

$$H_I = H_{DDEF} + H_{PPEF} \approx H - (H_{SQ} + H_0 + SYM - H \cdot \cos(\lambda)), \quad (4)$$

where H_{DDEF} and H_{PPEF} represent the magnetic fields associated with the disturbed dynamo electric fields and the prompt penetration electric fields, respectively.

In order to isolate H_{DDEF} and H_{PPEF} from each other, we proceed under the assumption that both ionospheric currents drive quasi-periodic fluctuations [Nishida, 1968a, Blanc and Richmond, 1980, Younas et al., 2020]. Hence, through a Fourier analysis, we apply pass-band filters for frequencies between $69.5 - 550\mu$Hz (corresponding to periods in 4 hours to 30 minutes range).

3.3 Total Electron Content

The overall ionosphere dynamics in each considered region were characterized by the vertical Total Electron Content (TEC) variations in the vicinity of the chosen magnetic OB. TEC values were derived from GNSS measurements using TayAbsTEC method Yasyukevich et al. [2015a,b], Mylnikova et al. [2015]. In particular, data from UCOE, LPAL, NPGJ and TSK2 receivers was used to obtain TEC values for TEO, GUI, JAI and KAK regions, respectively.

However, we have to consider that the open data obtained for this work, doesn't have the best quality, even more considering the period of time. The quality of data is a sum of errors which depend on the satellite swarm, so as the receptor itself. We could smooth some of these errors but couldn't be avoided at all. Considering also that we are studying results from points in different longitudinal sectors (Mexico, Canarias Islands, India and Japan), there can be issues by comparing absolute TEC values. To deal with this issue, we instead analyze relative values, which is, measuring TEC percentage difference with respect to the 27 days median TEC prior to the GS Vijaya Lekshmi et al. [2011], Sergeeva et al. [2017].

4 Results

In the following section, we focus on the results related to the geomagnetic response to the St. Patrick's GS for each OB listed in Table 1. Then we show the results of Equation 4 for isolating H_I, as well as the reconstruction of H_{PPEF} through Fourier analysis. Finally, we provide an analysis related to local time sector in which the GS developed as well as a comparison between each OB result.

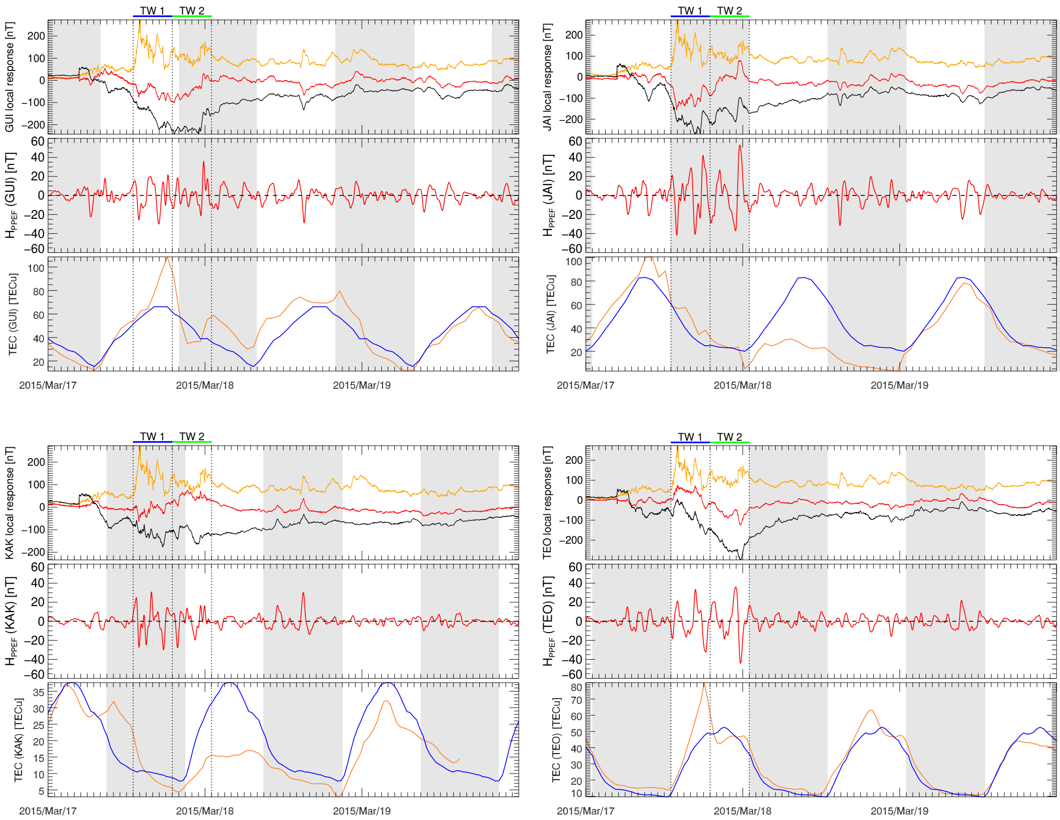

Figure 3: Top panels, local geomagnetic ΔH (black line), ASYH (yellow line) and the magnetic contribution of ionospheric origin (red line). Mid panels, reconstruction of H_{PPEF}. Bottom panels, TEC (orange line) and median TEC (blue line). In the upper side of each panel, the Time Windows (TW) are highlighted with a blue and black solid line respectively and vertical dotted lines. Each gray shading represents each night hours in Local Time

In Figure 3, we show in top panels, the local index ΔH (black line), the ASYH index (yellow line), and the computed ionospheric disturbance magnetic contribution H_I (red line) for each observatory. We focus on the period, TW, of high ASYH activity from 13:00 UT on March 17 to 01:00 UT the next day. As mentioned previously, this period is divided into two subsets (TW1 and TW2), highlighted by vertical dashed lines and two solid horizontal lines at the top of each panel (blue for TW1 and green for TW2). During these periods, H_I is primarily driven by H_{PPEF} fluctuations. The results of applying frequency band filters to the FFT(H_I), compared with the ionospheric response are shown in mid panels, represented by red lines. Finally, in bottom panels, we show the TEC response (orange lines), compared with the median TEC (blue line).

We will focus on periods where ASYH exhibited sharp temporal enhancements, or ASYH pulses, at different universal times. Our aim is to observe how each observatory registered different local responses depending on the local time at which these pulses occurred. The first pulse (273 nT at 14:07 UT), which is also the most intense, is our primary interest, but we identified three additional pulses. The second pulse (212 nT) occurs at 16:42 UT, while the third (150 nT) is located almost at the transition between TW1 and TW2, at 19:17 UT. Finally, we identify a fourth pulse (174 nT) around 23:00 UT, near the end of TW2.

For the ionospheric response, our primary interest is whether a positive or negative storm phase occurred, based on the available GNSS data used for vertical TEC computation. As mentioned previously, we analyze relative TEC variations with respect to a 27-day median. Since the observatories are at quite different longitudes, we define a significant disturbance as a TEC deviation exceeding 30%, following the threshold established by Vijaya Lekshmi et al. [2011].

We begin by analyzing the magnetic data from GUI. During its local TW1, GUI was on the dayside (11:00-17:00 LT), while TW2 occurred mostly on the nightside. The transition between TW1 and TW2 is nearly centered on the dusk sector (18:00 LT). The peak ASYH enhancement (274 nT) occurs at 12:07 LT at GUI, nearly coinciding with the noon sector. ΔH shows a smooth decrease during TW1, consistent with the main phase, while H_I appears to exhibit a pattern opposite to ASYH. During TW2, ASYH shows significantly reduced activity, with two smaller enhancements at 17:17 LT and 22:00 LT. In this period, ΔH maintains sustained minimum values around -212 nT until 21:00 LT, when it suddenly increases concurrently with an ASYH enhancement. H_{PPEF} fluctuation amplitudes within TW1 and TW2 (afternoon and night sectors, respectively) show peak amplitudes of -29.98 nT at 14:56 LT and 35.66 nT at 21:48 LT. Outside these peaks, we observe small contrast between H_{PPEF} amplitudes inside and outside the TW1 and TW2 periods. In TEC response, we observe an enhancement period around 16:00 LT, reaching an increment of around 55% of TECu, followed by a second enhancement of 58% almost at the end of TW2 and near midnight.

For JAI magnetic data, we observe that both time windows occur when JAI is on the night side. During TW1, the peak ASYH enhancement occurs at 19:07 LT, while both ΔH and H_I show a more pronounced decrease compared to GUI. ΔH reaches its minimum nearly simultaneously with the second ASYH pulse. During TW1, H_I appears to have a more clearly opposite pattern to ASYH, which changes completely during TW2. Within TW2 (post-midnight sector), ΔH exhibits periods of pronounced decreases and increases, differing from the behavior at GUI. Around 04:00 LT, there is a sharp increase in both ΔH and H_I, corresponding to the final ASYH enhancement in TW2. JAI observatory presents a higher contrast in H_{PPEF} fluctuations between inside and outside TW1 and TW2 (night and post-midnight sectors for JAI). The H_{PPEF} fluctuations show a sharp enhancement upon entering TW1, immediately after the dusk sector. During both TW1 and TW2, H_{PPEF} maintains values above $|20|$ nT, with peaks of -41.32 nT at 18:55 LT (TW1) and 52.73 nT at 04:55 LT (TW2). TEC response on the other hand, is does not present a significant differences until the end of TW2, on march 18. During this day, we observe a consistent negative response of around 70%.

At KAK, we observe during TW1 that ΔH shows a significantly less pronounced decrease during TW1 compared to the previous observatories. The minimum ΔH is reached at approximately 02:40 LT, temporally correlated with a small ASYH pulse. H_I also shows a pattern opposite to the ASYH index, though this is only evident until 01:00 LT. After this time, H_I gradually appears to follow a pattern more similar to ASYH. During TW2, H_I maintains intense negative values between -90 nT and -150 nT. There is no apparent response in ΔH during the final ASYH pulse beginning at 08:00 LT. Conversely, H_I

shows a growth phase between the two time windows, reaching a maximum value shortly after dawn at 06:00 LT. For KAK H_{PPEF} amplitudes are less intense than at JAI, with peaks of 30.58 nT at 00:48 LT and -29.78 nT at 02:40 LT, both within TW1. Apart from the observed contrast in H_{PPEF} amplitudes upon entering TW1, we observe a decreasing amplitude trend transitioning from TW1 to TW2 toward the dayside. In TEC, we observe similar to JAI, a relatively small enhancement which is consistent with the first stage of MP. Nevertheless, in this case this enhancement is presented during night hours, conforming bulb, or like a second peak. Next, there is a negative response of 50% the next day of MP.

For TEO, which represents the opposite case of JAI, we observe that both TW1 and TW2 occur during day-side hours (DWN and NDS sectors, respectively). In both time windows, we observe an almost linear descending trend in ΔH, unlike the previous cases where the behavior changed between time windows. ΔH reaches a minimum value at approximately 16:20 LT, consistent with the timing of the final ASYH pulse. H_I, on the other hand, shows a pattern more similar to ASYH behavior during TW1, with positive values that contrast with previous cases and peak values occurring approximately simultaneously with ASYH pulses. During TW2, however, H_I begins a decreasing phase, reaching a minimum at the same time as ΔH and the final ASYH enhancement. We observe H_{PPEF} amplitude peaks of 35.12 nT at 15:57 LT and -34.63 nT at 16:40 LT during TW2. TEO also shows a noticeable contrast in H_{PPEF} amplitudes inside versus outside the TW1 and TW2 periods. Examining the TEC relative response, we observe a sharp enhancement of 70% near the end of TW1 at 11:00 LT .

5 Discussion

From Figure 3, we observe significant variations in the results depending on the local time sector of each observatory during TW1 and TW2. More specifically, we note variations in the partial ring current (PRC) influence on the local response. When comparing the geomagnetic responses, we observe characteristic behaviors in ΔH and consequently in H_I.

For GUI, JAI, and KAK, a decrease in ΔH and consequently in H_I is observed during TW1, particularly at 14:07 UT (corresponding to 12:07, 19:07, and 23:07 LT, respectively). All three observatories exhibit a clear decreasing trend during TW1, with varying degrees of intensity. JAI shows the most pronounced decrease, while KAK displays the least pronounced.

TW2 also shows notable differences among the OB. At GUI, ΔH maintains sustained minimum values throughout most of TW2, while JAI and KAK exhibit both decreasing and increasing ΔH values. H_I shows an increasing trend at all three observatories until 23 UT (21, 04 and 08 LT). In contrast, TEO it is observed, after ASYH reaches its maximum at 07:07 LT, both ΔH and H_I show a consistent decreasing trend throughout TW1 and TW2 until the final ASYH pulse at approximately 23:00-00:00 UT (16:00-17:00 LT).

An interesting observation is that during various periods of TW1 and TW2, H_I sometimes appears to behave oppositely to ASYH activity (e.g., during TW1 at JAI), while in other cases it follows ASYH fluctuations more closely (e.g., during TW1 at TEO). In some instances, these similar or opposite patterns are only present during ASYH pulses of activity. This may imply the contribution of PRC to the Ionospheric response.

To quantitatively confirm the contribution of the PRC to H_I and consequently to H_{PPEF}, we conducted two validation experiments focusing on TW1 (13:00-19:00 UT) and TW2 (19:00-01:00 UT), during which PRC activity occurs. The experiments were:

1. Analyze the correlation between H_I and ASYH.
2. Analyze the H_{PPEF} fluctuations distribution during MP, looking for the sectors where H_{PPEF} is more intense in general.

In the first experiment, we computed the Pearson correlation coefficient (ρ) in two stages: first, ρ_1 for ASYH vs H_I, and second, ρ_2 for ASYH vs H_I after removing H_{PPEF} from H_I using a low-pass filter.

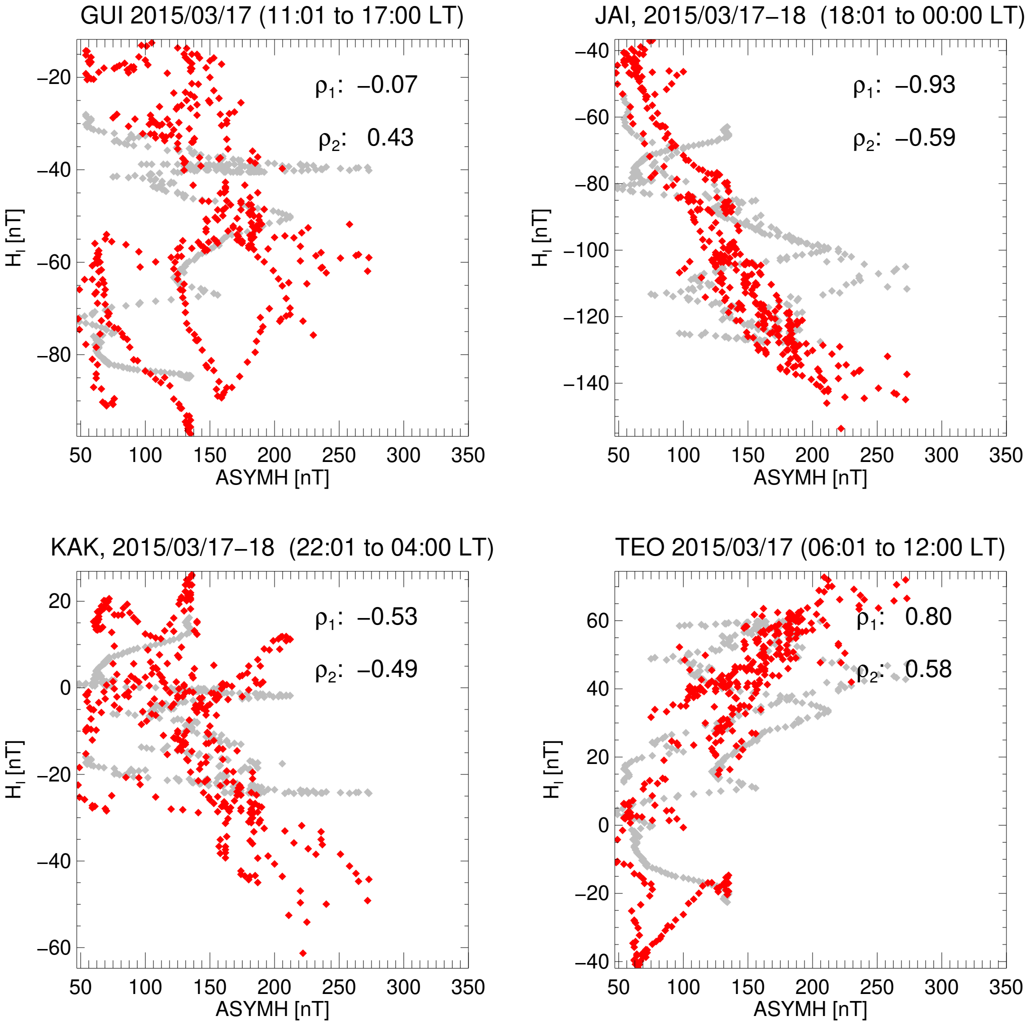

Figure 4: Correlation Analysis ASYH and H_I, during the period 1300 - 1900 UT, which is consistent with the period of highest ASYH activity

The results for TW1 are shown in Figure 4, where red diamonds represent ρ_1 (before removing H_{PPEF}), and gray diamonds represent ρ_2 (after removing H_{PPEF}).

According to Makarov [2022], it is implied that there is a strong anti-correlation between ASYH and SYMH indices. Therefore, a strongly positive or negative ρ coefficient can be interpreted as H_I (and consequently ΔH) being influenced by PRC activity, which is well described by the ASYH index. In this context, a strongly positive ρ implies a negative contribution to the local H component; in other words, the PRC may intensify the local geomagnetic response. Conversely, a strongly negative ρ suggests that the PRC may attenuate the local geomagnetic response.

Note that the two observatories showing strong correlation (or anti-correlation) are JAI and TEO. Interestingly, these were also the observatories closest to the dusk and dawn sectors, respectively, at the beginning of TW1. After removing H_{PPEF} (by a low-pass filter) from H_I, the scattering increases at all observatories except GUI. In case of correlation during TW2, the coefficients are: $\rho_{\text{GUI}} = 0.39$, $\rho_{\text{JAI}} = 0.19$, $\rho_{\text{KAK}} = -0.17$, and $\rho_{\text{TEO}} = -0.14$. Given that ASYH activity is significantly weaker during TW2, this suggests that ASYH intensity is an important factor influencing ΔH.

By measuring ρ during individual ASYH pulses, we find that the first pulse shows ρ values of [-0.81 (GUI), -0.91 (JAI), -0.70 (KAK), and 0.88 (TEO)]. For the second and third pulses, only JAI and TEO maintain correlations above 0.8 or below -0.8, while KAK shows values around 0.55. For the final ASYH pulse, only TEO exhibits a correlation slightly above 0.6 (or below -0.6). This supports the interpretation that H_{PPEF} magnitudes are strongly influenced by PRC activity, with a dependence on local time but also, the intensity of PRC activity being another factor for this influence.

Following the idea of PRC influencing on H_{PPEF} fluctuation amplitudes, we carried out an analysis of H_{PPEF} distribution magnitudes, from march 17^{th} to march 18^{th}, shown in Figure 5. From top panels, we observe that H_{PPEF} magnitudes (blue bars) in each OB seems to approach to a Gaussian distribution (fitted with a red line). All OB present media $\mu \sim 0$ nT and a very small skewness, seen at first glance. The standard deviation σ seen in each OB has a maximum difference goes from 5.2 to 8.15, being not so significant.

Of the four observatories, the KAK H_{PPEF} distribution is the narrowest and sharpest, characterized by the smallest standard deviation (σ), the highest peak, and the shortest tails. This indicates that KAK exhibits the least dispersed data with the fewest extreme values and the most stable H_{PPEF} amplitudes throughout the 2-day period. The opposite extreme is observed at JAI, which has the flattest and widest distribution, consistent with the largest σ, indicating the greatest overall data variability. JAI also possesses the longest tails, suggesting a significantly higher likelihood of extreme H_{PPEF} values.

The observatories GUI and TEO exhibit intermediate characteristics. GUI shows the second-largest dispersion after JAI, indicating highly variability in H_{PPEF} values. However, its tails are shorter than JAI's, meaning it has fewer extreme values than JAI despite its high variability. This high dispersion at GUI implies no strong systematic difference exists between the H_{PPEF} values in the specific time windows (TW1, TW2) and the entire 2-day period, a conclusion supported by the minor amplitude differences shown in Figure 3. In contrast, TEO's distribution is narrower than both GUI and JAI but exhibits a higher propensity for extreme values than both GUI and KAK. Consequently, the narrow distributions of KAK and TEO imply a more limited range of background H_{PPEF} amplitudes, making any deviations during the TW1 and TW2 windows more statistically significant relative to their low background variability.

From here, we are interested in estimating the point at which H_{PPEF} can be considered as extreme, for each OB. We refer this point as *Threshold*. Thus, higher thresholds imply higher extreme values in general.

First, from the good symmetry in all distributions, we can consider an absolute value threshold. From this approach, we set each threshold as the 95% of each cumulative H_{PPEF} probability distribution, shown in bottom panels of Figure 5 (blue lines). Since we are considering —H_{PPEF}—, we fit an accumulated half-Gaussian distribution (red lines). The threshold is indicated by the red dashed line for the 2 days period. We compare this threshold with the threshold of H_{PPEF} intensities within TW.

The highest couple of thresholds correspond to JAI (28.63 and 40.98 nT) followed by TEO (20.31 and 33.79). It is remarkable too how each threshold grew by considering only TW data. In JAI and TEO, the second threshold presents an increment of 43.13 and 66.37% respectively with respect to the first one. Meanwhile, in GUI and KAK, having significantly minor thresholds, the increment is about 35 and 45 % respectively. From here, we observe that the Threshold increment in KAK is not so different from JAI meanwhile TEO presents the highest threshold increment of the four OB. What this difference may indicate is how H_{PPEF} intensity is affected by the LT sector at the moment of the highest ASYH activity. This result is consistent with the results of $\rho_{1,2}$ for these OB, which reinforces the idea of H_{PPEF} being influenced by PRC activity, indicating not just the impact of PRC activity but also on the proximity of JAI and TEO to one of the joint points of PRC - FAC during its highest activity.

Regarding the ionospheric response, we observe that the TEO and GUI regions (monitored by the UCOE and LPAL receivers) present a positive storm effect, while the JAI and KAK regions (monitored by the NPGJ and TSK2 receivers) present a negative effect. From Thomas and Venables [1966], Vijaya Lekshmi et al. [2011], it is understood that positive phases at mid-low latitudes are often initiated on the day-side during the main phase of the storm. The physical mechanism for this is complex and involves thermospheric winds, $\mathbf{E} \times \mathbf{B}$ drift, and, most importantly for this study, the presence of eastward PPEF. According to Vijaya Lekshmi et al. [2011], the influence of PPEF alone may not always induce large TEC enhancements; its effects are often combined with neutral wind circulation and $\mathbf{E} \times \mathbf{B}$ drift to produce significant positive ionospheric storms.

The negative effects, on the other hand, are more related to regions that were on the night-side during the storm's main phase, which is the case for JAI and KAK in our event. These negative phases, however, typically manifest the following morning in local time. Studies like Fuller-Rowell et al. [1994], Prölss [1995] describe the mechanism responsible for these negative effects. It involves a storm-induced circulation that pushes molecular-rich air ([N_2], [O_2]) from high latitudes toward lower latitudes. This increases the $\frac{[N_2]}{[O]}$ ratio, which in turn enhances the loss rate of atomic oxygen ions (O^+) through recombination. Although this circulation begins during the night, its effects peak the next morning when photo-ionization commences but is overwhelmed by the heightened recombination rate, resulting in an electron density depletion and a negative TEC storm.

6 conclusions

Within this project, we analyzed longitudinal magnetic fluctuations driven by Prompt Penetration Electric Fields (PPEF) during the intense St. Patrick's Day geomagnetic storm on March 17, 2015. To achieve this, we utilized geomagnetic data from ground-based observatories separated by 60–90° in longitude.

The study focused on isolating the magnetic component associated with the ionospheric current system (H_I) and the fluctuations specifically linked to PPEF (H_{PPEF}). The H_{PPEF} signal was reconstructed via pass-band filtering to retain fluctuations with periods within re range of 4h and 30 minutes. Our analysis targeted the period of enhanced activity in the ASYH index, which reflects the asymmetric part of the magnetospheric ring current system and is linked to the Partial Ring Current (PRC).

We investigated the influence of the PRC on both H_I and H_{PPEF}. These observations were complemented by analyzing relative Vertical TEC variations, considering deviations above 30% from the 27-day median as significant ionospheric perturbations. We observed positive ionospheric storm effects at the TEO and GUI regions (55% and 70% TEC increase, respectively), while the JAI and KAK regions exhibited negative storm effects (70% and 50% TEC decrease). This pattern is consistent with the local time of each region during the storm's main phase onset, but also reflects the influence the impact of PPEF and neutral winds flow on electron density over those regions.

By comparing H_I and H_{PPEF} with ASYH activity across different local time sectors and integrating TEC observations, we reached the following conclusions:

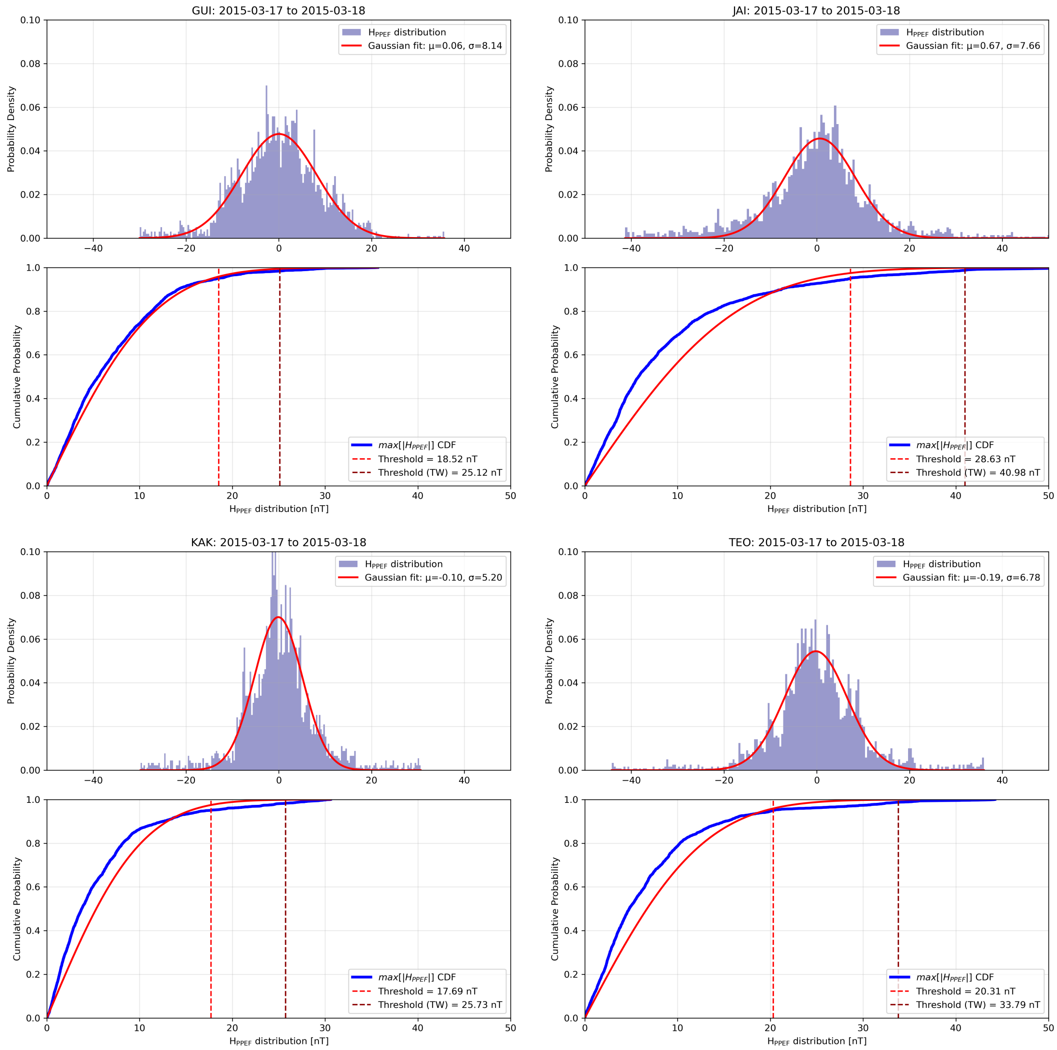

Figure 5: H_{PPEF} distribution for each OB. Color dashed vertical lines represent the a threshold for H_{PPEF} to be considered as intense amplitudes

- PRC activity is highly correlated with fluctuations in both H_I and H_{PPEF} during the most geoeffective period of the geomagnetic storm.
- The local geomagnetic response (ΔH) is either enhanced or attenuated depending on the observatory's proximity to the dawn or dusk sectors, where the PRC influence is strongest.
- High PRC activity significantly affects the magnitude of H_{PPEF}, particularly when an observatory is magnetically conjugate to the regions where the PRC connects to the ionosphere via Field-Aligned Currents (FACs).
- Conversely, when an observatory is located far from these PRC-FAC coupling regions (e.g., near the noon sector), the contribution to H_{PPEF} is less significant.
- The TEC enhancements at GUI and TEO are primarily driven by eastward PPEF combined with $\mathbf{E} \times \mathbf{B}$ drift from neutral wind circulation.
- The TEC depletions at JAI and KAK are attributed to storm-induced changes in thermospheric composition and circulation. The distinct mechanisms observed between the dayside and nightside regions suggest the presence of a disturbed dynamo.

7 Future work

This study constitutes the first part of a broader investigation into the local time dependence of the regional geomagnetic response. A subsequent study will focus on the effects of the Disturbance Dynamo. Furthermore, given the recent occurrence of intense geomagnetic storms, we plan to extend this analysis to events such as the "Mother's Day" storm and the event of October 10th. Expanding the dataset in this way will also help mitigate data quality issues, such as the limited availability of high-quality GNSS receiver data encountered in the present work.

A Processing magnetic data

It is crucial to eliminate regular variations from magnetic records to isolate magnetic contributions of interest. The main field H_0 can be removed by modeling a monthly baseline. This baseline was approximated by computing an array of daily mode values for March 2015, following Gjerloev [2012]. These daily mode values were obtained from data between 00:00 and 03:00 LT ($H_{0:3}$), then H_0 was computed by a linear least squares fitting from these daily values. To exclude data from disturbed days for the fitting, we adapted the procedure outlined in van de Kamp [2013]. Specifically, we applied a moving hourly window to calculate the hourly interquartile range (IQR). Disturbed days were identified by setting a variation threshold where:

$$IQR^i_{med} > threshold = disturbed \tag{5}$$

Here, IQR^i_{med} is the median of each daily IQR values, so if more than 50% if IQR values for a given day, surpass a determined threshold, then that day is considered as disturbed day and depreciated for the LS fitting.

The threshold was set, based on the criteria seen in Solari et al. [2017]. Basically, we analyze the monthly distribution of maximum IQR values (IQR^{max}_i) within tri-hours IQR bins, by shifting another moving window and sorting those peaks, so $IQR^{max}_1 < IQR^{max}_2 < ... < IQR^{max}_n$, in order to get a cumulative IQR^{max} distribution.

Hence, we have to compute the parameters of the General Pareto Distribution (GPD) Pickands [1975] which is a type of distribution used for modeling other distributions extreme values, and it is expressed as:

$$f(x) = \begin{cases} \frac{1}{\sigma}\left(1 + k\frac{x-u}{\sigma}\right)^{-1-\frac{1}{k}} & \text{if } k \neq 0 \\ \frac{1}{\sigma}exp\left(-\frac{x-u}{\sigma}\right) & \text{if } k = 0 \end{cases} \tag{6}$$

where the three GPD parameters are shape k, scale σ and location u for the cumulative IQR^{max}_i. The last parameter, u is the one which can be considered as a threshold point [Hosking, 1990, Solari et al., 2017]. Then, for each IQR_i^{max}, we compute $[k_i, \sigma_i, u_i]$, then we apply a modified Anderson-Darling test. The threshold, u_t, is considered once the test gives the most optimum validation result.

To compute the diurnal variation, we identified what we refer to as local quiet days (LQD) for each station. Following the methodology in van de Kamp [2013], we considered the five days with the smallest daily IQR peaks as LQDs. We then computed the average of the H component data during these five quietest days:

$$\langle LDQ_H(t)\rangle = \frac{1}{5}\sum_{i=1}^{5} H_i(t) - \langle H_i 0:3\rangle \tag{7}$$

where we normalize the offset of each daily data by subtracting the average of the first four hours of magnetic data in local time, as indicated by the second term in Equation 7. Subsequently, we model the diurnal variation H_{SQ} by applying a low-pass filter to the Fourier Transform of the result from Equation 7, following the procedure in Equation (1) of van de Kamp [2013].

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